Code file

for

Car number Plate Detection Model

Submitted by

Khushi Tomar

(Roll No: 25CSE1003000526)

Under the supervision of

Mr. Shobhit

IILM University, Greater Noida, Uttar Pradesh

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1. Uploading dataset taken from kaggle

```
+ Code   + Text

from google.colab import files
uploaded = files.upload()  # opens file chooser

Choose files  archive.zip
archive.zip(application/x-zip-compressed) - 252968601 bytes, last modified: 22/11/2025 - 100% done
Saving archive.zip to archive.zip
```

2. Unzip the file/dataset

```
!unzip -q archive.zip -d sample_data

!ls sample_data

anscombe.json                mnist_test.csv         YOLO_dataset
california_housing_test.csv  mnist_train_small.csv
california_housing_train.csv README.md
```

3. Install ultralytics for YOLOv8 model

```
[10]
5s
!pip install ultralytics

Collecting ultralytics
  Downloading ultralytics-8.3.230-py3-none-any.whl.metadata (37 kB)
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.12.0.88)
Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)
Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.9.0+cu126)
Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.24.0+cu126)
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.31.0)
Collecting ultralytics-thop>=2.0.18 (from ultralytics)
  Downloading ultralytics_thop-2.0.18-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (25.0)
```

4.Import yolov8 from ultralytics

```
[11]
4s
from ultralytics import YOLO

Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics.com/quickstart/#ultralytics
```

5. Update Datapath name

```
[20]
0s
# Update data.yaml with correct relative paths
import yaml

data_yaml_path = 'sample_data/YOLO_dataset/data.yaml'

with open(data_yaml_path, 'r') as f:
    data = yaml.safe_load(f)

# Assuming images are directly under YOLO_dataset/images/train and YOLO_dataset/images/val
# based on the standard YOLO dataset structure and typical unzipping behavior.
data['train'] = 'sample_data/YOLO_dataset/images/train'
data['val'] = 'sample_data/YOLO_dataset/images/val'

with open(data_yaml_path, 'w') as f:
    yaml.safe_dump(data, f)

print("data.yaml updated successfully with relative paths:")
with open(data_yaml_path, 'r') as f:
    print(f.read())

data.yaml updated successfully with relative paths:
names:
- number_plate
nc: 1
train: sample_data/YOLO_dataset/images/train
val: sample_data/YOLO_dataset/images/val
```

```
# Read the content of the data.yaml file to inspect its paths
with open('sample_data/YOLO_dataset/data.yaml', 'r') as f:
    data_yaml_content = f.read()

print(data_yaml_content)

train: /kaggle/input/license-plate-detection-dataset-anpr-yolo-format/images/train
val: /kaggle/input/license-plate-detection-dataset-anpr-yolo-format/images/val
nc: 1
names: ['number_plate']
```

6. Train the model using yolov8

```
[21]
✓ 11m
from ultralytics import YOLO
import yaml

# Define the path to your data.yaml file
data_yaml_path = "sample_data/YOLO_dataset/data.yaml"

# Load the existing data.yaml content
with open(data_yaml_path, 'r') as f:
    data = yaml.safe_load(f)

# Update the 'train' and 'val' paths to be relative to the data.yaml's location
data['train'] = 'images/train'
data['val'] = 'images/val'

# Save the corrected data.yaml content
with open(data_yaml_path, 'w') as f:
    yaml.safe_dump(data, f)

print("data.yaml updated successfully with corrected relative paths:")
with open(data_yaml_path, 'r') as f:
    print(f.read())

# Load YOLOv8 small model
model = YOLO("yolov8n.pt")  # pretrained base model

# Train the model with the corrected data.yaml
model.train(data=data_yaml_path, epochs=50, imgsz=640)

data.yaml updated successfully with corrected relative paths:
names:
```

Verifying model is trained

```
[22]
✓ 0s
!ls runs/detect

train  train2

[23]
✓ 0s
!ls runs/detect/train/weights
!ls runs/detect/train2/weights

best.pt  last.pt
```

7. Run the model

```
[25]
✓
from ultralytics import YOLO

model = YOLO("runs/detect/train2/weights/best.pt")  # or train/weights/best.pt
model.predict(source="Cars.png", save=True, conf=0.25)
```

8. Check the accuracy of the model

[26]
6s

```python
from ultralytics import YOLO

# Load your trained model
model = YOLO("runs/detect/train2/weights/best.pt")

# Validate the model
metrics = model.val(data="sample_data/YOLO_dataset/data.yaml")
```

```
Ultralytics 8.3.230 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 72 layers, 3,005,843 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 2258.6±1063.0 MB/s, size: 319.7 KB)
val: Scanning /content/sample_data/YOLO_dataset/labels/val.cache... 133 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 133/133 148.2Kit/s 0.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 9/9 2.5it/s 3.7s
                   all        133        145      0.892      0.917      0.937      0.597
Speed: 3.2ms preprocess, 10.8ms inference, 0.0ms loss, 2.4ms postprocess per image
Results saved to /content/runs/detect/val
```